

Smart Lockout 2.0 — a lockout / tagout portal that makes every padlock, permit and hand-over accountable.

A web application for plants that run lockout / tagout (LOTO) with smart padlock cabinets: a Work Permit Issuer isolates equipment and issues permits, a Work Permit Receiver signs on to do the work, and an administrator keeps users, equipment and cabinets in order. I designed the full portal — three role-based journeys, 40+ screens and the component set behind them.



Summary

Mission

Lockout / tagout is how a plant stops a boiler, panel or conveyor from being switched on while someone is working on it. In practice it runs on physical padlocks, paper permits and shift hand-overs — and every gap between those is a safety risk. Smart Lockout 2.0 connects the physical padlocks (iFobs held in smart cabinets) to a portal that tracks who isolated what, who holds which lock, which permits are open, and what happens when a shift ends or a permit expires. The product had to be usable by people in high-vis on a plant floor, not just by an admin at a desk.

Client

Loto Smart — Smart Lockout 2.0, a lockout / tagout and work-permit management portal for industrial plants.

What I did

- Role & journey mapping (WPI / WPR / Admin)
- Information architecture & dashboards
- Create Lockout with live cabinet status

My contributions

I mapped the three roles and their end-to-end journeys first — Work Permit Issuer, Work Permit Receiver and System Administrator — then designed every screen along them: sign-in, role dashboards, the create-lockout flow with live cabinet status, lockout and permit listings, sign-on / sign-off, adding WPRs and locks to an existing lockout, returning keys, transfers between shifts and task executors, expired-permit transfer, reports, and the admin side for user accounts and equipment. Along the way I built the dialog, table, status and card patterns the whole portal reuses.

Impact

Delivered as a complete high-fidelity Figma design — three role journeys, 40+ screens and 30+ dialog states — ready for development, with the same dashboard, table, dialog and status language across every role so a user who learns one flow can predict the rest.

- Lockout & permit listings
- Sign-on / sign-off & key return
- Shift, task-executor & expired-permit transfers
- Admin: users, equipment, alerts
- Reports & component patterns

My role

Product / UX Designer — sole designer, working with the product owner and engineering team.

Tools

Figma · FigJam

Timeline

2025

Three roles, one system of record.

The first job was to be precise about who does what. LOTO has a strict chain of responsibility, and the portal had to mirror it rather than flatten everyone into "user". Each role gets its own dashboard, with only the actions it is allowed to take.

ROLE 01

Work Permit Issuer (WPI)

Owens the isolation. Creates lockouts, picks the padlocks, issues permits, adds receivers and locks, signs off, and handles every kind of transfer.

- Create / change / sign off lockout
- Create permit, work-permit list
- Add WPR or lock to a lockout
- Return lock / key, clear status
- Transfer lockout, task executor, shift, expired permit
- Reports

ROLE 02

Work Permit Receiver (WPR)

Does the work. Signs on to a lockout, adds their own lock if allowed, signs on and off the permit, and hands over at shift change.

- View lockouts
- Sign on / sign off lockout
- Add lock to lockout
- Sign on / sign off work permit
- Transfer between shifts
- Return lock / key

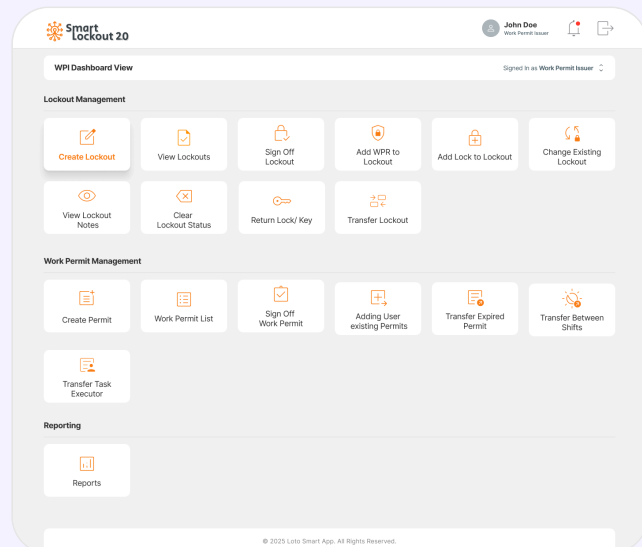
ROLE 03

System Administrator

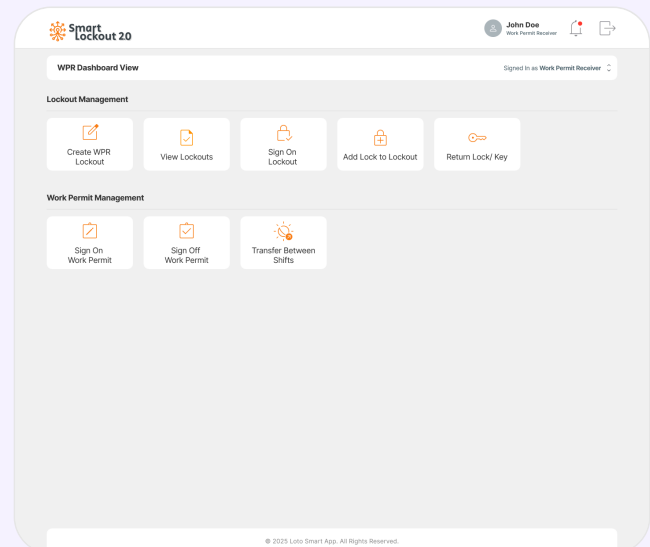
Keeps the setup healthy: people, equipment and cabinets. Sees cabinet and iFob errors first.

- Important alerts (cabinet / iFob / key errors)
- Create & update user accounts
- User account list
- Add equipment, equipment list
- Reports

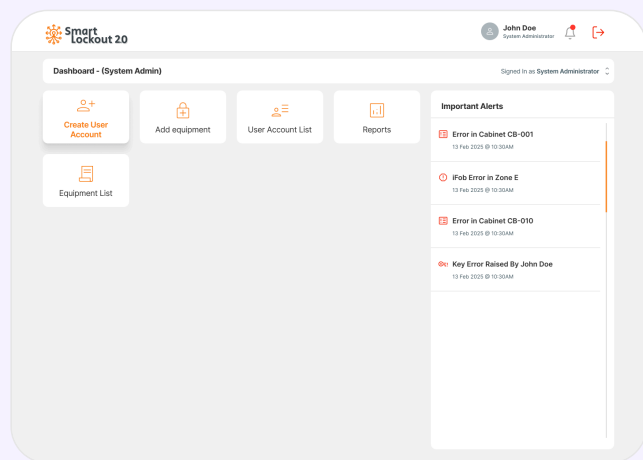
WPI dashboard — Lockout Management, Work Permit Management and Reporting as three groups of plain-language actions.



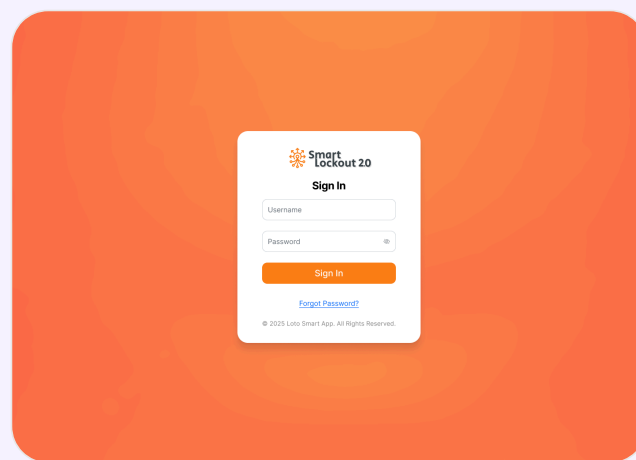
WPR dashboard — the same layout, a shorter list: only what a receiver is allowed to do.



Admin dashboard — Important Alerts sit above the setup actions, because a cabinet error matters more than a new account.



Sign in — one entry point; the role comes from the account and drives which dashboard opens.



DECISION

Actions as tiles, not menus

Every dashboard is a grid of large icon tiles with the action written out — "Sign Off Lockout", "Return Lock / Key". On a tablet at a cabinet, that beats a nested navigation.

DECISION

Say who you're signed in as

The role is printed in the header and repeated in the page bar ("Signed in as Work Permit Issuer"). In a safety system, nobody should have to guess whose permissions they're using.

Creating a lockout, with the cabinet in view.

The core flow. The issuer describes the equipment and the isolation reason on the left, and picks physical padlocks from the smart cabinet on the right – with a live count of what's available, in use and selected. The form can't be saved until iFobs are chosen, and the banner says what has to happen next: return the fobs to the cabinet before a receiver can be added.

01

Describe

Lockout ID is generated; equipment and isolation reason are required.

02

Pick padlocks

Choose iFobs from the cabinet grid, grouped ten at a time.

03

Authenticate

Card scan or password confirms the issuer; fobs go back to the cabinet.

04

Add WPR

Assign receivers and issue the work permit.

05

Sign off

Locks returned, status cleared, lockout complete.

Create Lockout — form on the left, cabinet A-123 on the right with Available / In Use / Selected counts.

Smart Lockout 2.0

John Doe
Work Permit Issuer

Create Lockout

Lockout ID: LO-2025-001

Equipment Description *
High-Pressure Boiler #23, Main Electrical Panel - Zone A

Isolation Reason *
Routine Maintenance, Emergency Shutdown

Selected iFobs *
Select iFob's from the list.

☐ Allow WPR to add additional padlocks to this lockout without WPI authentication

Notes
Enter extra notes

Please enter extra details if required.

Cancel

Save

iFobs Cabinet - A123

Select required iFob's (Padlocks) for the lockout.

Available: 24 In Use: 48 Selected: 0

Padlock 1 to 10

LO-001 LO-002 LO-003 LO-004 LO-005 LO-006

LO-007 LO-008 LO-009 LO-010

Padlock 11 to 20

LO-011 LO-012 LO-013 LO-014 LO-015 LO-016

LO-017 LO-018 LO-019 LO-020

View Lockouts — success banner, search, sortable table with Active / Pending / Complete status pills, permit count and row actions.

Smart Lockout 2.0

John Doe
Work Permit Issuer

View Lockouts

Create Lockout

Lockout Created Successfully! LO-2025-001 has been created. Before proceeding, please return all iFobs (padlocks) to the cabinet to complete the isolation process.

Search lockouts by ID, equip, or assigned user.

Lockout ID#	Status	WPI	Issue Date/Time (UTC)	Cabinet#	iFobs Padlocks	Equipment Description	No of Permits	Actions
LO-2025-001	Active	John Doe	Nov 24, 2025, 14:30	A-123	5	Conveyor Belt A1	-	
LO-2025-002	Active	Jane Roe	Nov 24, 2025, 15:00	B-112	5	Boiler System B2	3	
LO-2025-003	Pending	Bob Wilson	Nov 24, 2025, 15:45	C-221	4	Cooling Tower C3	1	
LO-2025-004	Active	Robert Brown	Nov 24, 2025, 17:10	A-222	5	Cooling System #4	2	
LO-2025-005	Complete	Jessica Hall	Nov 24, 2025, 18:45	B-112	5	Electrical Grid #9	-	
LO-2025-006	Active	Daniel Wilson	Nov 24, 2025, 18:45	A-007	2	Boiler #3	1	
LO-2025-007	Active	Emma Martinez	Nov 24, 2025, 15:45	C-010	2	Fuel System #6	1	
LO-2025-008	Pending	Sophia Carter	Nov 24, 2025, 15:45	B-315	3	Cooling Tower #8	1	
LO-2025-009	Active	Benjamin Evans	Nov 24, 2025, 15:45	A-420	1	Air Compressor #1	1	
LO-2025-010	Complete	Olivia Robinson	Nov 24, 2025, 15:45	C-221	5	Generator Panel #7	1	

Total 10 items

10 / page

Authentication Required — before a lockout is confirmed, the issuer proves it's them: RFID / NFC card scan or password. The same dialog frame carries every confirmation in the portal.

Authentication Required

Work Permit Issuer (WPI) must authenticate before proceeding.

Authentication methods:
Card Scan (RFID/NFC) or Password

Or

Enter Password

Proceed

Select Shifts — Morning / Day / Night as three big targets, with the current shift marked, before a hand-over.

Select Shifts

Morning

Day
Current Shift

Night

Cancel

Proceed

Abhishek Patade · abhipatade-creator.github.io/Portfolio2026

Smart Lockout 2.0 — Lockout / Tagout & Work Permit Portal · 5/9

DECISION

The cabinet is a picker, not a dropdown

Padlocks are shown as a grid that mirrors the physical cabinet, with a coloured dot for availability. Picking LO-006 on screen should feel like reaching for it on the wall.

DECISION

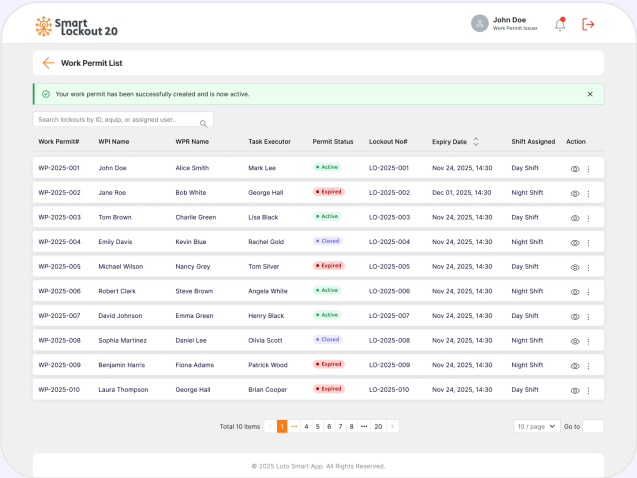
Rules live in banners, not in error messages

"A WPR can only be added once all iFobs are returned" is shown before the user tries, in an info banner — so the sequence is taught, not enforced by failure.

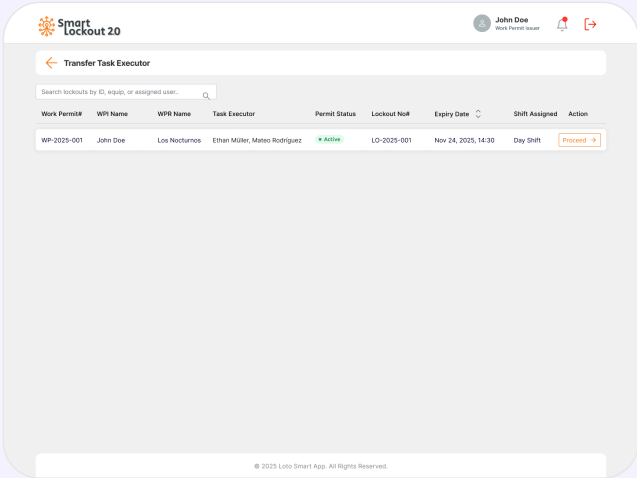
Permits, hand-overs and the messy middle.

Real plants don't finish work in one shift. The hardest screens were the transfers: moving a permit to a new task executor, handing a lockout between shifts, and dealing with a permit that expired while locks are still on. Each one is a listing with a Proceed action, a confirmation dialog, and a clear statement of what changes hands.

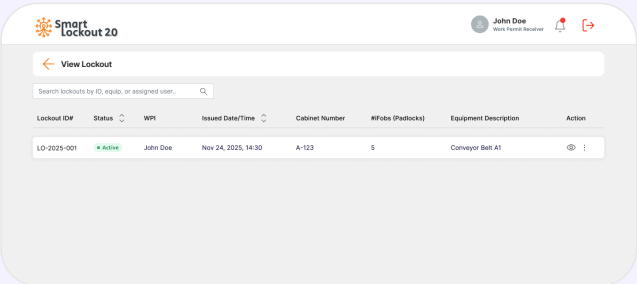
Work Permit List — permit number, WPI, WPR, task executor, Active / Expired / Closed status, linked lockout, expiry and shift in one row.



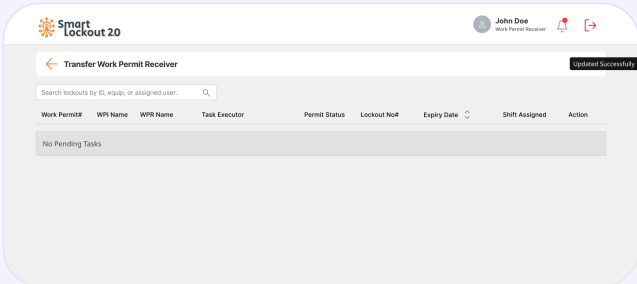
Transfer Task Executor — find the permit, see who currently holds it, Proceed to hand the work to someone else without touching the lockout.



WPR — View Lockouts. The receiver sees the lockouts they're on and signs on or off from the row.



WPR — Transfer Work Permit Receiver, including the empty state: "No Pending Tasks" is a full screen, not a blank table.



DECISION

One table pattern everywhere

Lockouts, permits and transfers all use the same table: search on top, status pills, a sortable date column, view and overflow actions on the right, pagination below. Four journeys, one thing to learn.

DECISION

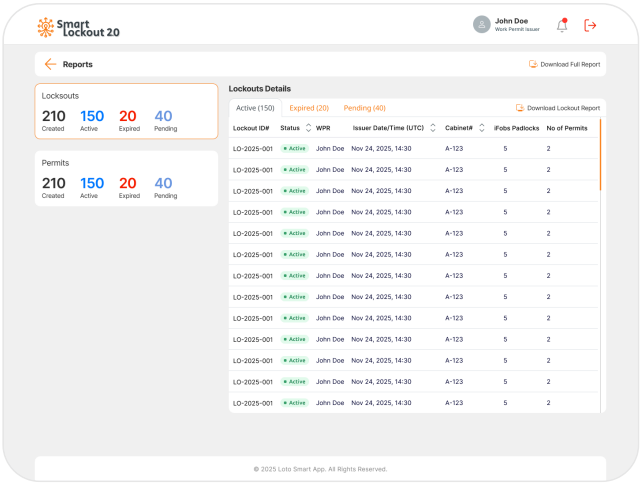
Transfers are a step, not a form

Every transfer starts from a listing and a single Proceed, then confirms in a dialog. The user picks a real row first — nobody types a permit number from memory.

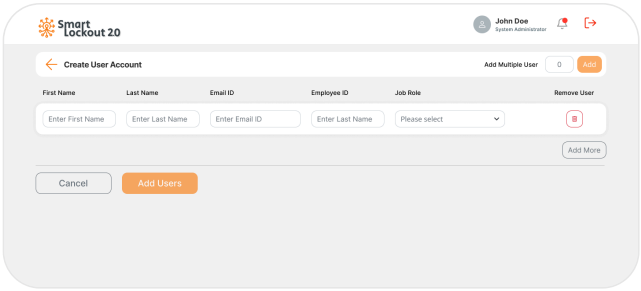
Admin and reporting.

The administrator's side is deliberately plain: create and update accounts with a role, register equipment, and keep the cabinets and iFobs healthy. Reports roll lockouts up into Created / Active / Expired / Pending counts with a downloadable detail table for each.

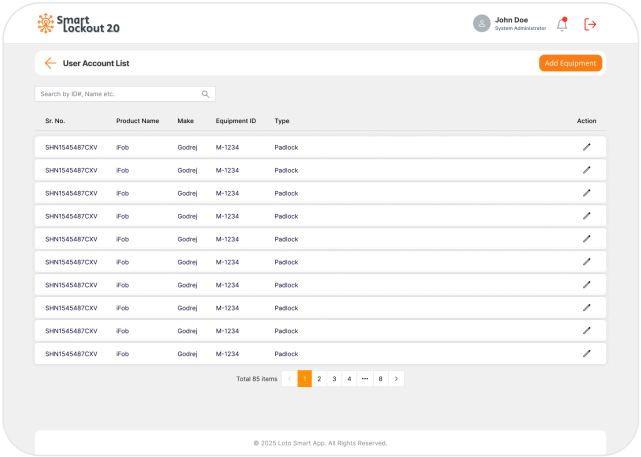
Reports (WPI) — headline counts, a filter by status, and the full lockout table ready to download.



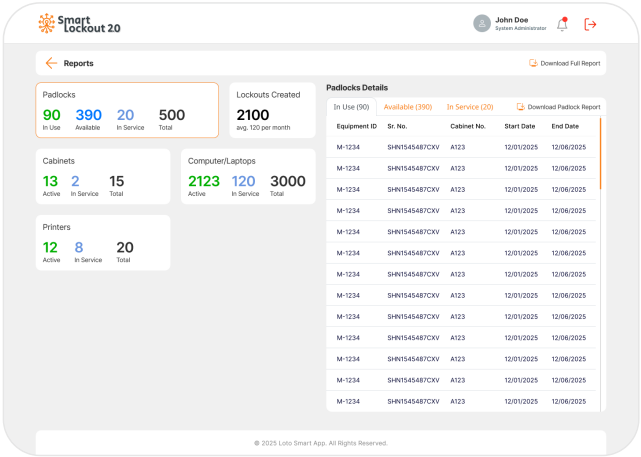
Create User Account — name, email, employee ID and role as a row; "Add multiple users" adds rows instead of opening another form.



Equipment list — every iFob and padlock registered in the system, with make, ID and type, and an Add Equipment action.



Admin reports — the same report frame as the issuer's view, scoped to the whole system.



What I'd carry into the next safety-critical product.

Map the roles before the screens.

Getting WPI, WPR and Admin right — who can do what, in what order — made forty screens fall out of three journeys instead of forty separate decisions.

Teach the rule before the error.

Info banners that explain sequencing up front removed a whole class of "why can't I save?" moments.

Mirror the physical world.

The padlock grid looks like the cabinet, the status colours match the dots on the fobs. When software describes a physical process, the mapping is the usability.

Design the exceptions.

Expired permits, shift changes, a receiver who leaves mid-job — the edge cases are where a safety system earns its keep, so they got full flows, not a footnote.